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JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 81 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let $\frac{x^2}{f(a^2+7a+3)}+\frac{y^2}{f(3a+15)}=1$ represent an ellipse with major axis along y-axis, where f is a strictly decreasing positive function on $\mathbf{R}$. If the set of all possible values of a is $\mathbf{R}-[\alpha,\beta]$, then $\alpha^2+\beta^2$ is equal to :

(A) 28 (B) 40 (C) 61 (D) 24

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $x=9$ be a directrix of an ellipse E , whose centre is at the origin and eccentricity is $\frac{1}{3}$. Let $\mathrm{P}(\alpha,0)$, $\alpha>0$, be a focus of E and AB be a chord passing through P . Then the locus of the mid point of AB is :

(A) $9y^2=8x(1-x)$ (B) $3y^2=4x(1-x)$ (C) $9y^2=8x(x-1)$ (D) $3y^2=4x(x-1)$

Open solution

Q3. JEE Main 2026 (Online) 5th April Morning Shift

Let a focus of the ellipse $\mathrm{E}:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ be $\mathrm{S}(4,0)$ and its eccentricity be $\frac{4}{5}$. If the point $\mathrm{P}(3,\alpha)$ lies on E and O is the origin, then the area of $\triangle\mathrm{POS}$ is equal to:

(A) 12/5 (B) 14/5 (C) 24/5 (D) 48/5

Open solution

Q4. JEE Main 2026 (Online) 4th April Evening Shift

Let $\mathrm{P}(3\cos\alpha,2\sin\alpha)$, $\alpha\neq0$, be a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, Q be a point on the circle $x^2+y^2-14x-14y+82=0$ and R be a point on the line $x+y=5$ such that the centroid of the triangle PQR is $\left(2+\cos\alpha,3+\frac{2}{3}\sin\alpha\right)$. Then the sum of the ordinates of all possible points R is:

(A) 6 (B) 2 (C) 4 (D) 8

Open solution

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

Let an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a<b$, pass through the point (4, 3) and have eccentricity $\frac{\sqrt{5}}{3}$.

Then the length of its latus rectum is :

(A) $\frac{4\sqrt{5}}{3}$ (B) $2\sqrt{5}$ (C) $\frac{7\sqrt{5}}{3}$ (D) $\frac{8\sqrt{5}}{3}$

Open solution

Q6. JEE Main 2026 (Online) 28th January Evening Shift

An ellipse has its center at $(1,-2)$, one focus at $(3,-2)$ and one vertex at $(5,-2)$. Then the length of its latus rectum is :

(A) 6 (B) $6\sqrt{3}$ (C) $\frac{16}{\sqrt{3}}$ (D) $4\sqrt{3}$

Open solution

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Q7. JEE Main 2026 (Online) 24th January Evening Shift

Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, be 30 . If its eccentricity is the maximum value of the function $f(t)=-\frac{3}{4}+2t-t^2$, then $\left(a^2+b^2\right)$ is equal to

(A) 276 (B) 516 (C) 256 (D) 496

Open solution

Q8. JEE Main 2026 (Online) 24th January Morning Shift

Let each of the two ellipses $\mathrm{E}_1: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$ and $\mathrm{E}_2: \frac{x^2}{\mathrm{A}^2}+\frac{y^2}{\mathrm{B}^2}=1,(\mathrm{A}<\mathrm{B})$ have eccentricity $\frac{4}{5}$. Let the lengths of the latus recta of E_1 and E_2 be l_1 and l_2, respectively, such that $2l_1^2=9l_2$. If the distance between the foci of E_1 is 8 , then the distance between the foci of E_2 is

(A) $\frac{96}{5}$ (B) $\frac{8}{5}$ (C) $\frac{16}{5}$ (D) $\frac{32}{5}$

Open solution

Q9. JEE Main 2026 (Online) 23rd January Evening Shift

If the points of intersection of the ellipses $x^2+2y^2-6x-12y+23=0$ and

$4x^2+2y^2-20x-12y+35=0$ lie on a circle of radius r and centre (a,b), then the

value of $ab+18r^2$ is :

(A) 53 (B) 52 (C) 55 (D) 51

Open solution

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

Let the line $y-x=1$ intersect the ellipse $\frac{x^2}{2}+\frac{y^2}{1}=1$ at the points A and B . Then the angle made by the line segment AB at the center of the ellipse is :

(A) $\pi-\tan^{-1}\left(\frac{1}{4}\right)$ (B) $\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{4}\right)$

(C) $\frac{\pi}{2}+2\tan^{-1}\left(\frac{1}{4}\right)$ (D) $\frac{\pi}{2}-\tan^{-1}\left(\frac{1}{4}\right)$

Open solution

Q11. JEE Main 2026 (Online) 22nd January Evening Shift

Let S and S′ be the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ and $\mathrm{P}(\alpha,\beta)$ be a point on the ellipse in the first quadrant. If $(\mathrm{SP})^2+(\mathrm{S'P})^2-\mathrm{SP}\cdot\mathrm{S'P}=37$, then $\alpha^2+\beta^2$ is equal to :

(A) 13 (B) 15 (C) 11 (D) 17

Open solution

Q12. JEE Main 2026 (Online) 21st January Evening Shift

If the line $\alpha x+4y=\sqrt{7}$, where $\alpha\in\mathbf{R}$, touches the ellipse $3x^2+4y^2=1$ at the point P in the first quadrant, then one of the focal distances of P is :

(A) $\frac{1}{\sqrt{3}}-\frac{1}{2\sqrt{11}}$ (B) $\frac{1}{\sqrt{3}}-\frac{1}{2\sqrt{5}}$

(C) $\frac{1}{\sqrt{3}}+\frac{1}{2\sqrt{5}}$ (D) $\frac{1}{\sqrt{3}}+\frac{1}{2\sqrt{7}}$

Open solution

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Q13. JEE Main 2025 (Online) 8th April Evening Shift

Let the ellipse $3x^2 + py^2 = 4$ pass through the centre C of the circle $x^2 + y^2 - 2x - 4y - 11 = 0$ of radius r. Let f_1, f_2 be the focal distances of the point C on the ellipse. Then $6f_1f_2 - r$ is equal to

(A) 78 (B) 68 (C) 70 (D) 74

Open solution

Q14. JEE Main 2025 (Online) 7th April Evening Shift

Let the length of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be 10. If its eccentricity is the minimum value of the function $f(t) = t^2 + t + \frac{11}{12}, t \in \mathbb{R}$, then $a^2 + b^2$ is equal to :

(A) 115 (B) 120 (C) 125 (D) 126

Open solution

Q15. JEE Main 2025 (Online) 7th April Evening Shift

Let p be the number of all triangles that can be formed by joining the vertices of a regular polygon P of n sides and q be the number of all quadrilaterals that can be formed by joining the vertices of P. If p + q = 126, then the eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{n} = 1$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{3}{4}$

Open solution

Q16. JEE Main 2025 (Online) 4th April Evening Shift

Let for two distinct values of p the lines $y = x + \mathrm{p}$ touch the ellipse $\mathrm{E} : \frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$ at the points A and B . Let the line $y = x$ intersect E at the points C and D . Then the area of the quadrilateral $ABCD$ is equal to :

(A) 48 (B) 20 (C) 24 (D) 36

Open solution

Q17. JEE Main 2025 (Online) 4th April Evening Shift

The centre of a circle C is at the centre of the ellipse $\mathrm{E} : \frac{x^2}{\mathrm{a}^2} + \frac{y^2}{\mathrm{b}^2} = 1, \mathrm{a} > \mathrm{b}$. Let C pass through the foci F_1 and F_2 of E such that the circle C and the ellipse E intersect at four points. Let P be one of these four points. If the area of the triangle $\mathrm{PF}_1\mathrm{F}_2$ is 30 and the length of the major axis of E is 17 , then the distance between the foci of E is :

(A) 12 (B) 26 (C) 13 (D) $\frac{13}{2}$

Open solution

Q18. JEE Main 2025 (Online) 4th April Morning Shift

The length of the latus-rectum of the ellipse, whose foci are $(2, 5)$ and $(2, -3)$ and eccentricity is $\frac{4}{5}$, is

(A) $\frac{50}{3}$ (B) $\frac{18}{5}$ (C) $\frac{6}{5}$ (D) $\frac{10}{3}$

Open solution

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Q19. JEE Main 2025 (Online) 3rd April Evening Shift

Let C be the circle of minimum area enclosing the ellipse $E: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with eccentricity $\frac{1}{2}$ and foci $(\pm 2, 0)$. Let PQR be a variable triangle, whose vertex P is on the circle C and the side QR of length $2a$ is parallel to the major axis of E and contains the point of intersection of E with the negative y-axis. Then the maximum area of the triangle PQR is :

(A) $8(3+\sqrt{2})$ (B) $8(2+\sqrt{3})$ (C) $6(3+\sqrt{2})$ (D) $6(2+\sqrt{3})$

Open solution

Q20. JEE Main 2025 (Online) 3rd April Morning Shift

A line passing through the point $P(\sqrt{5}, \sqrt{5})$ intersects the ellipse $\frac{x^2}{36}+\frac{y^2}{25}=1$ at A and B such that $(PA)\cdot(PB)$ is maximum. Then $5\left(PA^2+PB^2\right)$ is equal to :

(A) 290 (B) 377 (C) 338 (D) 218

Open solution

Q21. JEE Main 2025 (Online) 2nd April Evening Shift

If the length of the minor axis of an ellipse is equal to one fourth of the distance between the foci, then the eccentricity of the ellipse is :

(A) $\frac{3}{\sqrt{19}}$ (B) $\frac{\sqrt{3}}{16}$ (C) $\frac{4}{\sqrt{17}}$ (D) $\frac{\sqrt{5}}{7}$

Open solution

Q22. JEE Main 2025 (Online) 2nd April Morning Shift

If S and S' are the foci of the ellipse $\frac{x^2}{18}+\frac{y^2}{9}=1$ and P be a point on the ellipse, then $\min(SP\cdot S'P)+\max(SP\cdot S'P)$ is equal to :

(A) $3(6+\sqrt{2})$ (B) $3(1+\sqrt{2})$ (C) 27 (D) 9

Open solution

Q23. JEE Main 2025 (Online) 29th January Evening Shift

If $\alpha x+\beta y=109$ is the equation of the chord of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, whose mid point is $\left(\frac{5}{2}, \frac{1}{2}\right)$. then $\alpha+\beta$ is equal to :

(A) 37 (B) 46 (C) 72 (D) 58

Open solution

Q24. JEE Main 2025 (Online) 29th January Morning Shift

Let the ellipse $E_1: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$ and $E_2: \frac{x^2}{A^2}+\frac{y^2}{B^2}=1, A<B$ have same eccentricity $\frac{1}{\sqrt{3}}$. Let the product of their lengths of latus rectums be $\frac{32}{\sqrt{3}}$ and the distance between the foci of E_1 be 4. If E_1 and E_2 meet at A, B, C and D, then the area of the quadrilateral ABCD equals :

(A) $\frac{24\sqrt{6}}{5}$ (B) $\frac{18\sqrt{6}}{5}$ (C) $6\sqrt{6}$ (D) $\frac{12\sqrt{6}}{5}$

Open solution

Q25. JEE Main 2025 (Online) 28th January Evening Shift

If the midpoint of a chord of the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is $(\sqrt{2}, 4/3)$, and the length of the chord is $\frac{2\sqrt{\alpha}}{3}$, then α is :

(A) 26 (B) 18 (C) 22 (D) 20

Open solution

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Q26. JEE Main 2025 (Online) 24th January Evening Shift

The equation of the chord, of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, whose mid-point is $(3,1)$ is :

(A) $5x+16y=31$ (B) $48x+25y=169$ (C) $4x+122y=134$ (D) $25x+101y=176$

Open solution

Q27. JEE Main 2025 (Online) 24th January Morning Shift

Let the product of the focal distances of the point $\left(\sqrt{3},\frac{1}{2}\right)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, be $\frac{7}{4}$. Then the absolute difference of the eccentricities of two such ellipses is

(A) $\frac{1-2\sqrt{2}}{\sqrt{3}}$ (B) $\frac{1-\sqrt{3}}{\sqrt{2}}$ (C) $\frac{3-2\sqrt{2}}{2\sqrt{3}}$ (D) $\frac{3-2\sqrt{2}}{3\sqrt{2}}$

Open solution

Q28. JEE Main 2025 (Online) 23rd January Evening Shift

The length of the chord of the ellipse $\frac{x^2}{4}+\frac{y^2}{2}=1$, whose mid-point is $\left(1,\frac{1}{2}\right)$, is :

(A) $\frac{2}{3}\sqrt{15}$ (B) $\frac{1}{3}\sqrt{15}$ (C) $\sqrt{15}$ (D) $\frac{5}{3}\sqrt{15}$

Open solution

Q29. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\mathrm{E}:\frac{x^2}{\mathrm{a}^2}+\frac{y^2}{\mathrm{b}^2}=1,\mathrm{a}>\mathrm{b}$ and $\mathrm{H}:\frac{x^2}{\mathrm{A}^2}-\frac{y^2}{\mathrm{B}^2}=1$. Let the distance between the foci of E and the foci of H be $2\sqrt{3}$. If $a-A=2$, and the ratio of the eccentricities of E and H is $\frac{1}{3}$, then the sum of the lengths of their latus rectums is equal to :

(A) 10 (B) 7 (C) 9 (D) 8

Open solution

Q30. JEE Main 2024 (Online) 9th April Morning Shift

Let $f(x)=x^2+9, g(x)=\frac{x}{x-9}$ and $\mathrm{a}=f\circ g(10), \mathrm{b}=g\circ f(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{\mathrm{a}}+\frac{y^2}{\mathrm{b}}=1$, then $8\mathrm{e}^2+l^2$ is equal to.

(A) 6 (B) 12 (C) 8 (D) 16

Open solution

Q31. JEE Main 2024 (Online) 5th April Morning Shift

Let the line $2x+3y-\mathrm{k}=0,\mathrm{k}>0$, intersect the x-axis and y-axis at the points A and B, respectively. If the equation of the circle having the line segment AB as a diameter is $x^2+y^2-3x-2y=0$ and the length of the latus rectum of the ellipse $x^2+9y^2=k^2$ is $\frac{m}{n}$, where m and n are coprime, then $2\,\mathrm{m}+\mathrm{n}$ is equal to

(A) 12 (B) 13 (C) 11 (D) 10

Open solution

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Q32. JEE Main 2024 (Online) 1st February Evening Shift

Let P be a point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$. Let the line passing through P and parallel to y-axis meet the circle $x^2+y^2=9$ at point Q such that P and Q are on the same side of the x-axis. Then, the eccentricity of the locus of the point R on PQ such that $PR : RQ = 4 : 3$ as P moves on the ellipse, is :

(A) $\frac{13}{21}$ (B) $\frac{\sqrt{139}}{23}$ (C) $\frac{\sqrt{13}}{7}$ (D) $\frac{11}{19}$

Open solution

Q33. JEE Main 2024 (Online) 1st February Morning Shift

Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \mathrm{a}>\mathrm{b}$ be an ellipse, whose eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is $\sqrt{14}$. Then the square of the eccentricity of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is :

(A) 3 (B) $\frac{7}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Open solution

Q34. JEE Main 2024 (Online) 31st January Evening Shift

Let P be a parabola with vertex $(2,3)$ and directrix $2x+y=6$. Let an ellipse $E : \frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$, of eccentricity $\frac{1}{\sqrt{2}}$ pass through the focus of the parabola P. Then, the square of the length of the latus rectum of E, is

(A) $\frac{512}{25}$ (B) $\frac{656}{25}$ (C) $\frac{385}{8}$ (D) $\frac{347}{8}$

Open solution

Q35. JEE Main 2024 (Online) 30th January Evening Shift

Let $A(\alpha,0)$ and $B(0,\beta)$ be the points on the line $5x+7y=50$. Let the point P divide the line segment AB internally in the ratio $7:3$. Let $3x-25=0$ be a directrix of the ellipse $E : \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the corresponding focus be S. If from S, the perpendicular on the x-axis passes through P, then the length of the latus rectum of E is equal to,

(A) $\frac{25}{3}$ (B) $\frac{25}{9}$ (C) $\frac{32}{5}$ (D) $\frac{32}{9}$

Open solution

Q36. JEE Main 2024 (Online) 30th January Morning Shift

If the length of the minor axis of an ellipse is equal to half of the distance between the foci, then the eccentricity of the ellipse is :

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{5}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{5}}{3}$

Open solution

Q37. JEE Main 2024 (Online) 27th January Morning Shift

The length of the chord of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, whose mid point is $\left(1,\frac{2}{5}\right)$, is equal to :

(A) $\frac{\sqrt{1691}}{5}$ (B) $\frac{\sqrt{2009}}{5}$ (C) $\frac{\sqrt{1541}}{5}$ (D) $\frac{\sqrt{1741}}{5}$

Open solution

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Q38. JEE Main 2023 (Online) 12th April Morning Shift

Let $\mathrm{P}\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right), \mathrm{Q}, \mathrm{R}$ and S be four points on the ellipse $9x^2 + 4y^2 = 36$. Let PQ and RS be mutually perpendicular and pass through the origin. If $\frac{1}{(PQ)^2} + \frac{1}{(RS)^2} = \frac{p}{q}$, where p and q are coprime, then $p + q$ is equal to :

Open solution

(A) 143 (B) 147 (C) 137 (D) 157

Q39. JEE Main 2023 (Online) 11th April Morning Shift

Consider ellipses $\mathrm{E}_k : kx^2 + k^2y^2 = 1, k = 1, 2, \ldots, 20$. Let C_k be the circle which touches the four chords joining the end points (one on minor axis and another on major axis) of the ellipse E_k. If r_k is the radius of the circle C_k, then the value of $\sum_{k=1}^{20} \frac{1}{r_k^2}$ is :

Open solution

(A) 2870 (B) 3210 (C) 3320 (D) 3080

Q40. JEE Main 2023 (Online) 10th April Morning Shift

Let the ellipse $E : x^2 + 9y^2 = 9$ intersect the positive x and y-axes at the points A and B respectively. Let the major axis of E be a diameter of the circle C. Let the line passing through A and B meet the circle C at the point P. If the area of the triangle with vertices A, P and the origin O is $\frac{m}{n}$, where m and n are coprime, then $m - n$ is equal to :

Open solution

(A) 15 (B) 16 (C) 17 (D) 18

Q41. JEE Main 2023 (Online) 6th April Evening Shift

In a group of 100 persons 75 speak English and 40 speak Hindi. Each person speaks at least one of the two languages. If the number of persons, who speak only English is α and the number of persons who speak only Hindi is β, then the eccentricity of the ellipse $25\left(\beta^2x^2 + \alpha^2y^2\right) = \alpha^2\beta^2$ is :

Open solution

(A) $\frac{\sqrt{129}}{12}$ (B) $\frac{3\sqrt{15}}{12}$ (C) $\frac{\sqrt{119}}{12}$ (D) $\frac{\sqrt{117}}{12}$

Q42. JEE Main 2022 (Online) 29th July Morning Shift

Let a line L pass through the point of intersection of the lines $bx + 10y - 8 = 0$ and $2x - 3y = 0$, $\mathrm{b} \in \mathbf{R} - \left\{\frac{4}{3}\right\}$. If the line L also passes through the point $(1, 1)$ and touches the circle $17\left(x^2 + y^2\right) = 16$, then the eccentricity of the ellipse $\frac{x^2}{5} + \frac{y^2}{\mathrm{b}^2} = 1$ is :

Open solution

(A) $\frac{2}{\sqrt{5}}$ (B) $\sqrt{\frac{3}{5}}$ (C) $\frac{1}{\sqrt{5}}$ (D) $\sqrt{\frac{2}{5}}$

Q43. JEE Main 2022 (Online) 25th July Evening Shift

If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the line $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the eccentricity of the ellipse is :

Open solution

(A) $\frac{5}{7}$ (B) $\frac{2\sqrt{6}}{7}$ (C) $\frac{3}{7}$ (D) $\frac{2\sqrt{5}}{7}$

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Q44. JEE Main 2022 (Online) 30th June Morning Shift

Let the eccentricity of the ellipse $x^2 + a^2y^2 = 25a^2$ be b times the eccentricity of the hyperbola $x^2 - a^2y^2 = 5$, where a is the minimum distance between the curves y = e^x and y = $\log_e x$. Then $a^2 + \frac{1}{b^2}$ is equal to :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) 3 (D) 5

Open solution

Q45. JEE Main 2022 (Online) 27th June Morning Shift

Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$, be $\frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$, then $a^2 + b^2$ is equal to :

(A) 29 (B) 31 (C) 32 (D) 34

Open solution

Q46. JEE Main 2022 (Online) 26th June Evening Shift

The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Open solution

Q47. JEE Main 2022 (Online) 25th June Evening Shift

The line y = x + 1 meets the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$ at two points P and Q. If r is the radius of the circle with PQ as diameter then $(3r)^2$ is equal to :

(A) 20 (B) 12 (C) 11 (D) 8

Open solution

Q48. JEE Main 2022 (Online) 24th June Evening Shift

Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1, a > 2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{4}$

Open solution

Q49. JEE Main 2021 (Online) 31st August Evening Shift

The locus of mid-points of the line segments joining $(-3, -5)$ and the points on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is :

(A) $9x^2 + 4y^2 + 18x + 8y + 145 = 0$ (B) $36x^2 + 16y^2 + 90x + 56y + 145 = 0$

(C) $36x^2 + 16y^2 + 108x + 80y + 145 = 0$ (D) $36x^2 + 16y^2 + 72x + 32y + 145 = 0$

Open solution

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Q50. JEE Main 2021 (Online) 27th August Morning Shift

If $x^2 + 9y^2 - 4x + 3 = 0$, x, y $\in$ R, then x and y respectively lie in the intervals :

(A) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

(B) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ and [1, 3]

(C) [1, 3] and [1, 3]

(D) [1, 3] and $\left[-\frac{1}{3}, \frac{1}{3}\right]$

Open solution

Q51. JEE Main 2021 (Online) 27th July Morning Shift

A ray of light through (2, 1) is reflected at a point P on the y-axis and then passes through the point (5, 3). If this reflected ray is the directrix of an ellipse with eccentricity $\frac{1}{3}$ and the distance of the nearer focus from this directrix is $\frac{8}{\sqrt{53}}$, then the equation of the other directrix can be :

(A) $11x + 7y + 8 = 0$ or $11x + 7y - 15 = 0$

(B) $11x - 7y - 8 = 0$ or $11x + 7y + 15 = 0$

(C) $2x - 7y + 29 = 0$ or $2x - 7y - 7 = 0$

(D) $2x - 7y - 39 = 0$ or $2x - 7y - 7 = 0$

Open solution

Q52. JEE Main 2021 (Online) 25th July Morning Shift

Let an ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$, passes through $\left(\sqrt{\frac{3}{2}}, 1\right)$ and has eccentricity $\frac{1}{\sqrt{3}}$. If a circle, centered at focus F(α, 0), $\alpha > 0$, of E and radius $\frac{2}{\sqrt{3}}$, intersects E at two points P and Q, then PQ^2 is equal to :

(A) $\frac{8}{3}$ (B) $\frac{4}{3}$ (C) $\frac{16}{3}$ (D) 3

Open solution

Q53. JEE Main 2021 (Online) 22th July Evening Shift

Let $E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$. Let E_2 be another ellipse such that it touches the end points of major axis of E_1 and the foci of E_2 are the end points of minor axis of E_1. If E_1 and E_2 have same eccentricities, then its value is :

(A) $\frac{-1+\sqrt{5}}{2}$ (B) $\frac{-1+\sqrt{8}}{2}$ (C) $\frac{-1+\sqrt{3}}{2}$ (D) $\frac{-1+\sqrt{6}}{2}$

Open solution

Q54. JEE Main 2021 (Online) 25th February Evening Shift

If the curve $x^2 + 2y^2 = 2$ intersects the line x + y = 1 at two points P and Q, then the angle subtended by the line segment PQ at the origin is :

(A) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$ (B) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{3}\right)$

(C) $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{3}\right)$ (D) $\frac{\pi}{2} + \tan^{-1}\left(\frac{1}{4}\right)$

Open solution

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Q55. JEE Main 2020 (Online) 5th September Morning Slot

If the co-ordinates of two points A and B are $\left(\sqrt{7}, 0\right)$ and $\left(-\sqrt{7}, 0\right)$ respectively and P is any point on the conic, $9x^2 + 16y^2 = 144$, then PA + PB is equal to :

Open solution

(A) 8 (B) 9 (C) 16 (D) 6

Q56. JEE Main 2020 (Online) 4th September Morning Slot

Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a > b) be a given ellipse, length of whose latus rectum is 10. If its eccentricity is the maximum value of the function, $\phi(t) = \frac{5}{12} + t - t^2$, then $a^2 + b^2$ is equal to :

Open solution

(A) 145 (B) 126 (C) 135 (D) 116

Q57. JEE Main 2020 (Online) 7th January Morning Slot

If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is :

Open solution

(A) $\sqrt{3}$ (B) $3\sqrt{2}$ (C) $\frac{3}{\sqrt{2}}$ (D) $2\sqrt{3}$

Q58. JEE Main 2019 (Online) 12th April Evening Slot

An ellipse, with foci at (0, 2) and (0, −2) and minor axis of length 4, passes through which of the following points?

Open solution

(A) $\left(2, \sqrt{2}\right)$ (B) $\left(2, 2\sqrt{2}\right)$ (C) $\left(\sqrt{2}, 2\right)$ (D) $\left(1, 2\sqrt{2}\right)$

Q59. JEE Main 2019 (Online) 8th April Evening Slot

In an ellipse, with centre at the origin, if the difference of the lengths of major axis and minor axis is 10 and one of the foci is at $(0, 5\sqrt{3})$, then the length of its latus rectum is :

Open solution

(A) 5 (B) 8 (C) 10 (D) 6

Q60. JEE Main 2019 (Online) 12th January Evening Slot

Let S and S' be the foci of an ellipse and B be any one of the extremities of its minor axis. If $\Delta S'BS$ is a right angled triangle with right angle at B and area ($\Delta S'BS$) = 8 sq. units, then the length of a latus rectum of the ellipse is :

Open solution

(A) 2 (B) $4\sqrt{2}$ (C) 4 (D) $2\sqrt{2}$

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Q61. JEE Main 2019 (Online) 11th January Evening Slot

Let the length of the latus rectum of an ellipse with its major axis along x-axis and centre at the origin, be 8. If the distance between the foci of this ellipse is equal to the length of its minor axis, then which one of the following points lies on it?

(A) $\left(4\sqrt{2}, 2\sqrt{3}\right)$ (B) $\left(4\sqrt{3}, 2\sqrt{3}\right)$

(C) $\left(4\sqrt{3}, 2\sqrt{2}\right)$ (D) $\left(4\sqrt{2}, 2\sqrt{2}\right)$

Open solution

Q62. JEE Main 2019 (Online) 10th January Evening Slot

Let S = $\left\{(x, y) \in R^2 : \frac{y^2}{1+r} - \frac{x^2}{1-r} = 1\right\}; r \neq \pm 1$. Then S represents :

(A) an ellipse whose eccentricity is $\frac{1}{\sqrt{r+1}}$, where r > 1

(B) an ellipse whose eccentricity is $\frac{2}{\sqrt{r+1}}$, where 0 < r < 1

(C) an ellipse whose eccentricity is $\frac{2}{\sqrt{r-1}}$, where 0 < r < 1

(D) an ellipse whose eccentricity is $\sqrt{\frac{2}{r+1}}$, where r > 1

Open solution

Q63. JEE Main 2018 (Online) 16th April Morning Slot

If the length of the latus rectum of an ellipse is 4 units and the distance between a focus and its nearest vertex on the major axis is $\frac{3}{2}$ units, then its eccentricity is :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{9}$

Open solution

Q64. JEE Main 2017 (Online) 9th April Morning Slot

The eccentricity of an ellipse having centre at the origin, axes along the co-ordinate axes and passing through the points (4, −1) and (−2, 2) is :

(A) $\frac{1}{2}$ (B) $\frac{2}{\sqrt{5}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{4}$

Open solution

Q65. JEE Main 2017 (Online) 8th April Morning Slot

Consider an ellipse, whose center is at the origin and its major axis is along the x-axis. If its eccentricity is $\frac{3}{5}$ and the distance between its foci is 6, then the area (in sq. units) of the quadrilatateral inscribed in the ellipse, with the vertices as the vertices of the ellipse, is :

(A) 8 (B) 32 (C) 80 (D) 40

Open solution

Q66. JEE Main 2015 (Online) 11th April Morning Slot

If the distance between the foci of an ellipse is half the length of its latus rectum, then the eccentricity of the ellipse is :

(A) $\frac{1}{2}$ (B) $\frac{2\sqrt{2}-1}{2}$ (C) $\sqrt{2}-1$ (D) $\frac{\sqrt{2}-1}{2}$

Open solution

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Q67. JEE Main 2015 (Online) 10th April Morning Slot

An ellipse passes through the foci of the hyperbola, $9x^2 - 4y^2 = 36$ and its major and minor axes lie along the transverse and conjugate axes of the hyperbola respectively. If the product of eccentricities of the two conics is $\frac{1}{2}$, then which of the following points does not lie on the ellipse ?

(A) $\left(\sqrt{13}, 0\right)$
(B) $\left(\frac{\sqrt{39}}{2}, \sqrt{3}\right)$
(C) $\left(\frac{1}{2}\sqrt{13}, \frac{\sqrt{3}}{2}\right)$
(D) $\left(\sqrt{\frac{13}{2}}, \sqrt{6}\right)$

Open solution

Q68. JEE Main 2014 (Online) 11th April Morning Slot

A stair-case of length l rests against a vertical wall and a floor of a room. Let P be a point on the stair-case, nearer to its end on the wall, that divides its length in the ratio 1 : 2. If the stair-case begins to slide on the floor, then the locus of P is :

(A) an ellipse of eccentricity $\frac{1}{2}$
(B) an ellipse of eccentricity $\frac{\sqrt{3}}{2}$
(C) a circle of radius $\frac{l}{2}$
(D) a circle of radius $\frac{\sqrt{3}}{2}l$

Open solution

Q69. JEE Main 2014 (Online) 9th April Morning Slot

If OB is the semi-minor axis of an ellipse, F_1 and F_2 are its foci and the angle between F_1 B and F_2 B is a right angle, then the square of the eccentricity of the ellipse is :

(A) $\frac{1}{2}$
(B) $\frac{1}{\sqrt{2}}$
(C) $\frac{1}{2\sqrt{2}}$
(D) $\frac{1}{4}$

Open solution

Q70. JEE Main 2014 (Offline)

The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is :

(A) $\left(x^2 + y^2\right)^2 = 6x^2 + 2y^2$
(B) $\left(x^2 + y^2\right)^2 = 6x^2 - 2y^2$
(C) $\left(x^2 - y^2\right)^2 = 6x^2 + 2y^2$
(D) $\left(x^2 - y^2\right)^2 = 6x^2 - 2y^2$

Open solution

Q71. JEE Main 2013 (Online) 22th April Morning Slot

Let the equations of two ellipses be
E_1: $\frac{x^2}{3} + \frac{y^2}{2} = 1$ and
E_2: $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$
If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 is

(A) 8
(B) 9
(C) 4
(D) 2

Open solution

Q72. JEE Main 2013 (Online) 9th April Morning Slot

Equation of the line passing through the points of intersection of the parabola $x^2 = 8y$ and the ellipse $\frac{x^2}{3} + y^2 = 1$ is:

(A) $y - 3 = 0$
(B) $y + 3 = 0$
(C) $3y + 1 = 0$
(D) $3y - 1 = 0$

Open solution

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Q73. JEE Main 2013 (Online) 9th April Morning Slot

If a and c are positive real numbers and the ellipse $\frac{x^2}{4c^2}+\frac{y^2}{c^2}=1$ has four distinct points in common with the circle $x^2+y^2=9a^2$, then

(A) $9ac-9a^2-2c^2<0$ (B) $6ac+9a^2-2c^2<0$

(C) $9ac-9a^2-2c^2>0$ (D) $6ac+9a^2-2c^2>0$

Open solution

Q74. JEE Main 2013 (Offline)

The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, and having centre at $(0,3)$ is :

(A) $x^2+y^2-6y-7=0$ (B) $x^2+y^2-6y+7=0$

(C) $x^2+y^2-6y-5=0$ (D) $x^2+y^2-6y+5=0$

Open solution

Q75. AIEEE 2012

An ellipse is drawn by taking a diameter of thec circle $(x-1)^2+y^2=1$ as its semi-minor axis and a diameter of the circle $x^2+(y-2)^2=4$ is semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is :

(A) $4x^2+y^2=4$ (B) $x^2+4y^2=8$ (C) $4x^2+y^2=8$ (D) $x^2+4y^2=16$

Open solution

Q76. AIEEE 2011

Equation of the ellipse whose axes are the axes of coordinates and which passes through the point $(-3,1)$ and has eccentricity $\sqrt{\frac{2}{5}}$ is :

(A) $5x^2+3y^2-48=0$ (B) $3x^2+5y^2-15=0$

(C) $5x^2+3y^2-32=0$ (D) $3x^2+5y^2-32=0$

Open solution

Q77. AIEEE 2009

The ellipse $x^2+4y^2=4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point $(4,0)$. Then the equation of the ellipse is :

(A) $x^2+12y^2=16$ (B) $4x^2+48y^2=48$ (C) $4x^2+64y^2=48$ (D) $x^2+16y^2=16$

Open solution

Q78. AIEEE 2008

A focus of an ellipse is at the origin. The directrix is the line $x=4$ and the eccentricity is $\frac{1}{2}$. Then the length of the semi-major axis is :

(A) $\frac{8}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{5}{3}$

Open solution

Q79. AIEEE 2006

In the ellipse, the distance between its foci is 6 and minor axis is 8. Then its eccentricity is :

(A) $\frac{3}{5}$ (B) $\frac{1}{2}$ (C) $\frac{4}{5}$ (D) $\frac{1}{\sqrt{5}}$

Open solution

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Q80. AIEEE 2005

An ellipse has OB as semi minor axis, F and F' its foci and the angle FBF' is a right angle. Then the eccentricity of the ellipse is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{\sqrt{3}}$

Open solution

Q81. AIEEE 2004

The eccentricity of an ellipse, with its centre at the origin, is $\frac{1}{2}$. If one of the directrices is $x = 4$, then the equation of the ellipse is :

(A) $4x^2 + 3y^2 = 1$ (B) $3x^2 + 4y^2 = 12$ (C) $4x^2 + 3y^2 = 12$ (D) $3x^2 + 4y^2 = 1$

Open solution

Numerical 7 questions

Q82. JEE Main 2026 (Online) 4th April Morning Shift

Consider the parabola $\mathrm{P} : y^2 = 4kx$ and the ellipse $\mathrm{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let the line segment joining the points of intersection of P and E , be their latus rectums. If the eccentricity of E is e, then $e^2 + 2\sqrt{2}$ is equal to ____ .

Open solution

Q83. JEE Main 2026 (Online) 2nd April Evening Shift

Let A be the point (3, 0) and circles with variable diameter AB touch the circle $x^2 + y^2 = 36$ internally. Let the curve C be the locus of the point B. If the eccentricity of C is e, then $72e^2$ is equal to _____.

Open solution

Q84. JEE Main 2026 (Online) 24th January Evening Shift

Let (h, k) lie on the circle $\mathrm{C} : x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e. Then the value of $\frac{5}{e^2}$ is equal to ____ .

Open solution

Q85. JEE Main 2025 (Online) 28th January Morning Shift

Let $\mathrm{E}_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ be an ellipse. Ellipses E_i 's are constructed such that their centres and eccentricities are same as that of E_1, and the length of minor axis of E_i is the length of major axis of $E_{i+1} (i \geq 1)$. If A_i is the area of the ellipse E_i, then $\frac{5}{\pi}\left(\sum_{i=1}^{\infty} A_i\right)$, is equal to _____.

Open solution

Q86. JEE Main 2023 (Online) 15th April Morning Shift

Let an ellipse with centre $(1, 0)$ and latus rectum of length $\frac{1}{2}$ have its major axis along x-axis. If its minor axis subtends an angle 60° at the foci, then the square of the sum of the lengths of its minor and major axes is equal to _____.

Open solution

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Q87. JEE Main 2023 (Online) 24th January Morning Shift

Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C, then $10\,\alpha^2$ is equal to _____

Open solution

Q88. JEE Main 2022 (Online) 27th July Morning Shift

If the length of the latus rectum of the ellipse $x^2 + 4y^2 + 2x + 8y - \lambda = 0$ is 4 , and l is the length of its major axis, then $\lambda + l$ is equal to _____.

Open solution

Out of syllabus 41 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q89. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

Let the tangent and normal at the point $(3\sqrt{3}, 1)$ on the ellipse $\frac{x^2}{36} + \frac{y^2}{4} = 1$ meet the y-axis at the points A and B respectively. Let the circle C be drawn taking AB as a diameter and the line $x = 2\sqrt{5}$ intersect C at the points P and Q. If the tangents at the points P and Q on the circle intersect at the point (α, β), then $\alpha^2 - \beta^2$ is equal to :

(A) 61 (B) $\frac{304}{5}$ (C) 60 (D) $\frac{314}{5}$

Open solution

Q90. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

If the radius of the largest circle with centre (2,0) inscribed in the ellipse $x^2 + 4y^2 = 36$ is r, then $12r^2$ is equal to :

(A) 72 (B) 92 (C) 115 (D) 69

Open solution

Q91. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let a circle of radius 4 be concentric to the ellipse $15x^2 + 19y^2 = 285$. Then the common tangents are inclined to the minor axis of the ellipse at the angle :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Open solution

Q92. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

If the maximum distance of normal to the ellipse $\frac{x^2}{4} + \frac{y^2}{b^2} = 1, b < 2$, from the origin is 1, then the eccentricity of the ellipse is :

(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{2}$

Open solution

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Q93. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point $(1, 3)$ is :

(A) $\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$ (B) $\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

(C) $\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$ (D) $\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Open solution

Q94. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If m is the slope of a common tangent to the curves $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $x^2 + y^2 = 12$, then $12m^2$ is equal to :

(A) 6 (B) 9 (C) 10 (D) 12

Open solution

Q95. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Let θ be the acute angle between the tangents to the ellipse $\frac{x^2}{9} + \frac{y^2}{1} = 1$ and the circle $x^2 + y^2 = 3$ at their point of intersection in the first quadrant. Then $\tan\theta$ is equal to :

(A) $\frac{5}{2\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{4}{\sqrt{3}}$ (D) 2

Open solution

Q96. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

An angle of intersection of the curves, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $x^2 + y^2 = ab$, $a > b$, is :

(A) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$ (B) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$

(C) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$ (D) $\tan^{-1}\left(2\sqrt{ab}\right)$

Open solution

Q97. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

The line $12x\cos\theta + 5y\sin\theta = 60$ is tangent to which of the following curves?

(A) $x^2 + y^2 = 169$ (B) $144x^2 + 25y^2 = 3600$

(C) $25x^2 + 12y^2 = 3600$ (D) $x^2 + y^2 = 60$

Open solution

Q98. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

On the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$ let P be a point in the second quadrant such that the tangent at P to the ellipse is perpendicular to the line $x + 2y = 0$. Let S and S' be the foci of the ellipse and e be its eccentricity. If A is the area of the triangle SPS' then, the value of $(5 - e^2)$. A is :

(A) 6 (B) 12 (C) 14 (D) 24

Open solution

Q99. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

If a tangent to the ellipse $x^2 + 4y^2 = 4$ meets the tangents at the extremities of it major axis at B and C, then the circle with BC as diameter passes through the point :

(A) $(\sqrt{3}, 0)$ (B) $(\sqrt{2}, 0)$ (C) $(1, 1)$ (D) $(-1, 1)$

Open solution

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Q100. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

Let a tangent be drawn to the ellipse $\frac{x^2}{27} + y^2 = 1$ at $(3\sqrt{3}\cos\theta, \sin\theta)$ where $\theta \in \left(0, \frac{\pi}{2}\right)$. Then the value of θ such that the sum of intercepts on axes made by this tangent is minimum is equal to :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{8}$ (D) $\frac{\pi}{4}$

Open solution

Q101. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

If the points of intersections of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = 4b$, $b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to :

(A) 12 (B) 10 (C) 6 (D) 5

Open solution

Q102. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

If the normal at an end of a latus rectum of an ellipse passes through an extremity of the minor axis, then the eccentricity e of the ellipse satisfies :

(A) $e^4 + 2e^2 - 1 = 0$ (B) $e^4 + e^2 - 1 = 0$ (C) $e^2 + 2e - 1 = 0$ (D) $e^2 + e - 1 = 0$

Open solution

Q103. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

Which of the following points lies on the locus of the foot of perpendicular drawn upon any tangent to the ellipse,
$\frac{x^2}{4} + \frac{y^2}{2} = 1$
from any of its foci?

(A) $\left(-1, \sqrt{3}\right)$ (B) $\left(-2, \sqrt{3}\right)$ (C) $\left(-1, \sqrt{2}\right)$ (D) $(1, 2)$

Open solution

Q104. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

Let x = 4 be a directrix to an ellipse whose centre is at the origin and its eccentricity is $\frac{1}{2}$. If P(1, β), $\beta > 0$ is a point on this ellipse, then the equation of the normal to it at P is :

(A) $4x - 3y = 2$ (B) $8x - 2y = 5$ (C) $7x - 4y = 1$ (D) $4x - 2y = 1$

Open solution

Q105. JEE Main 2020 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The length of the minor axis (along y-axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, x + 6y = 8; then its eccentricity is :

(A) $\frac{1}{3}\sqrt{\frac{11}{3}}$ (B) $\frac{1}{2}\sqrt{\frac{5}{3}}$

(C) $\sqrt{\frac{5}{6}}$ (D) $\frac{1}{2}\sqrt{\frac{11}{3}}$

Open solution

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Q106. JEE Main 2020 (Online) 8th January Morning Slot · MCQ (Single Correct Answer)

Let the line y = mx and the ellipse $2x^2 + y^2 = 1$ intersect at a point P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}}, 0\right)$ and $(0, \beta)$, then β is equal to :

(A) $\frac{\sqrt{2}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Open solution

Q107. JEE Main 2020 (Online) 7th January Evening Slot · MCQ (Single Correct Answer)

If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ for some $a \in R$, then the distance between the foci of the ellipse is :

(A) $2\sqrt{5}$ (B) $2\sqrt{7}$ (C) 4 (D) $2\sqrt{2}$

Open solution

Q108. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the normal to the ellipse $3x^2 + 4y^2 = 12$ at a point P on it is parallel to the line, $2x + y = 4$ and the tangent to the ellipse at P passes through Q(4,4) then PQ is equal to :

(A) $\frac{\sqrt{61}}{2}$ (B) $\frac{\sqrt{221}}{2}$ (C) $\frac{\sqrt{157}}{2}$ (D) $\frac{5\sqrt{5}}{2}$

Open solution

Q109. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

The tangent and normal to the ellipse $3x^2 + 5y^2 = 32$ at the point P(2, 2) meet the x-axis at Q and R, respectively. Then the area (in sq. units) of the triangle PQR is :

(A) $\frac{14}{3}$ (B) $\frac{16}{3}$ (C) $\frac{68}{15}$ (D) $\frac{34}{15}$

Open solution

Q110. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the line x – 2y = 12 is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $\left(3, -\frac{9}{2}\right)$, then the length of the latus rectum of the ellipse is :

(A) 5 (B) 9 (C) $8\sqrt{3}$ (D) $12\sqrt{2}$

Open solution

Q111. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

If the tangent to the parabola $y^2 = x$ at a point (α, β), $(\beta > 0)$ is also a tangent to the ellipse, $x^2 + 2y^2 = 1$, then α is equal to :

(A) $\sqrt{2} + 1$ (B) $\sqrt{2} - 1$ (C) $2\sqrt{2} + 1$ (D) $2\sqrt{2} - 1$

Open solution

Q112. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

If the tangents on the ellipse $4x^2 + y^2 = 8$ at the points (1, 2) and (a, b) are perpendicular to each other, then a^2 is equal to :

(A) $\frac{2}{17}$ (B) $\frac{64}{17}$ (C) $\frac{128}{17}$ (D) $\frac{4}{17}$

Open solution

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Q113. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ at all points on the ellipse other than its four vertices then the mid points of the tangents intercepted between the coordinate axes lie on the curve :

(A) $\frac{x^2}{2} + \frac{y^2}{4} = 1$ (B) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$

(C) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$ (D) $\frac{x^2}{4} + \frac{y^2}{2} = 1$

Open solution

Q114. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

If β is one of the angles between the normals to the ellipse, $x^2 + 3y^2 = 9$ at the points $(3\cos\theta, \sqrt{3}\sin\theta)$ and $(-3\sin\theta, \sqrt{3}\cos\theta)$; $\theta \in \left(0, \frac{\pi}{2}\right)$; then $\frac{2\cot\beta}{\sin 2\theta}$ is equal to :

(A) $\frac{2}{\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{3}}{4}$

Open solution

Q115. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The eccentricity of an ellipse whose centre is at the origin is $\frac{1}{2}$. If one of its directrices is $x = -4$, then the equation of the normal to it at $\left(1, \frac{3}{2}\right)$ is :

(A) $2y - x = 2$ (B) $4x - 2y = 1$ (C) $4x + 2y = 7$ (D) $x + 2y = 4$

Open solution

Q116. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the tangent at a point on the ellipse $\frac{x^2}{27} + \frac{y^2}{3} = 1$ meets the coordinate axes at A and B, and O is the origin, then the minimum area (in sq. units) of the triangle OAB is :

(A) $\frac{9}{2}$ (B) $3\sqrt{3}$ (C) $9\sqrt{3}$ (D) 9

Open solution

Q117. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$, is :

(A) $\frac{27}{2}$ (B) 27 (C) $\frac{27}{4}$ (D) 18

Open solution

Q118. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

The minimum area of a triangle formed by any tangent to the ellipse $\frac{x^2}{16} + \frac{y^2}{81} = 1$ and the co-ordinate axes is :

(A) 12 (B) 18 (C) 26 (D) 36

Open solution

Q119. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

A point on the ellipse, $4x^2 + 9y^2 = 36$, where the normal is parallel to the line, $4x - 2y - 5 = 0$

(A) $\left(\frac{9}{5}, \frac{8}{5}\right)$ (B) $\left(\frac{8}{5}, -\frac{9}{5}\right)$

(C) $\left(-\frac{9}{5}, \frac{8}{5}\right)$ (D) $\left(\frac{8}{5}, \frac{9}{5}\right)$

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Q120. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The normal at $\left(2, \frac{3}{2}\right)$ to the ellipse, $\frac{x^2}{16} + \frac{y^2}{3} = 1$ touches a parabola, whose equation is

(A) $y^2 = -104x$ (B) $y^2 = 14x$ (C) $y^2 = 26x$ (D) $y^2 = -14x$

Open solution

Q121. AIEEE 2012 (Online) 12th May · MCQ (Single Correct Answer)

If P_1 and P_2 are two points on the ellipse $\frac{x^2}{4} + y^2 = 1$ at which the tangents are parallel to the chord joining the points $(0, 1)$ and $(2, 0)$, then the distance between P_1 and P_2 is

(A) $2\sqrt{2}$ (B) $\sqrt{5}$ (C) $2\sqrt{3}$ (D) $\sqrt{10}$

Open solution

Q122. AIEEE 2012 · MCQ (Single Correct Answer)

STATEMENT-1 : An equation of a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$ is $y = 2x + 2\sqrt{3}$

STATEMENT-2 : If line $y = mx + \frac{4\sqrt{3}}{m}, (m \neq 0)$ is a common tangent to the parabola $y^2 = 16\sqrt{3}x$ and the ellipse $2x^2 + y^2 = 4$, then m satisfies $m^4 + 2m^2 = 24$

(A) Statement-1 is false, Statement-2 is true.

(B) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(C) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(D) Statement-1 is true, Statement-2 is false.

Open solution

Q123. JEE Main 2023 (Online) 1st February Evening Shift · Numerical

The line $x = 8$ is the directrix of the ellipse $\mathrm{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the corresponding focus $(2, 0)$. If the tangent to E at the point P in the first quadrant passes through the point $(0, 4\sqrt{3})$ and intersects the x-axis at Q, then $(3\mathrm{PQ})^2$ is equal to _____.

Open solution

Q124. JEE Main 2023 (Online) 24th January Morning Shift · Numerical

Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersect the coordinate axes at the points A and B. Then, the minimum length of the line segment AB is _____

Open solution

Q125. JEE Main 2022 (Online) 28th July Evening Shift · Numerical

Let the tangents at the points P and Q on the ellipse $\frac{x^2}{2} + \frac{y^2}{4} = 1$ meet at the point $R(\sqrt{2}, 2\sqrt{2} - 2)$. If S is the focus of the ellipse on its negative major axis, then $\mathrm{SP}^2 + \mathrm{SQ}^2$ is equal to _____.

Open solution

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Q126. JEE Main 2022 (Online) 24th June Morning Shift · **Numerical**

If two tangents drawn from a point (α, β) lying on the ellipse $25x^2 + 4y^2 = 1$ to the parabola $y^2 = 4x$ are such that the slope of one tangent is four times the other, then the value of $(10\alpha + 5)^2 + (16\beta^2 + 50)^2$ equals _____.

Open solution

Q127. JEE Main 2021 (Online) 27th August Morning Shift · **Numerical**

If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2} + \frac{y^2}{4a^2} = 1$ and the co-ordinate axis is kab, then k is equal to _____.

Open solution

Q128. JEE Main 2021 (Online) 27th July Evening Shift · **Numerical**

Let E be an ellipse whose axes are parallel to the co-ordinates axes, having its center at $(3, -4)$, one focus at $(4, -4)$ and one vertex at $(5, -4)$. If $mx - y = 4$, $m > 0$ is a tangent to the ellipse E, then the value of $5m^2$ is equal to _____.

Open solution

Q129. JEE Main 2021 (Online) 26th February Evening Shift · **Numerical**

Let L be a common tangent line to the curves

$4x^2 + 9y^2 = 36$ and $(2x)^2 + (2y)^2 = 31$. Then the

square of the slope of the line L is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **C**	4. **D**	5. **D**
6. **A**	7. **D**	8. **D**	9. **C**	10. **B**
11. **A**	12. **D**	13. **C**	14. **D**	15. **A**
16. **C**	17. **C**	18. **B**	19. **B**	20. **C**
21. **C**	22. **C**	23. **D**	24. **A**	25. **C**
26. **B**	27. **C**	28. **A**	29. **D**	30. **C**
31. **C**	32. **C**	33. **C**	34. **B**	35. **C**
36. **B**	37. **A**	38. **D**	39. **D**	40. **C**
41. **C**	42. **B**	43. **A**	44. **D**	45. **B**
46. **C**	47. **A**	48. **A**	49. **C**	50. **D**
51. **C**	52. **C**	53. **A**	54. **D**	55. **A**
56. **B**	57. **B**	58. **C**	59. **A**	60. **C**
61. **C**	62. **D**	63. **B**	64. **C**	65. **D**
66. **C**	67. **C**	68. **B**	69. **A**	70. **A**
71. **C**	72. **D**	73. **C**	74. **A**	75. **D**
76. **D**	77. **A**	78. **A**	79. **A**	80. **A**
81. **B**				

Numerical

82. **3**	83. **18**	84. **9**	85. **54**	86. **9**
87. **118**	88. **75**			

Out of syllabus

89. **B**	90. **B**	91. **B**	92. **D**	93. **B**
94. **B**	95. **B**	96. **C**	97. **B**	98. **A**
99. **A**	100. **A**	101. **A**	102. **B**	103. **A**
104. **D**	105. **D**	106. **A**	107. **B**	108. **D**
109. **C**	110. **B**	111. **A**	112. **A**	113. **B**
114. **A**	115. **B**	116. **D**	117. **B**	118. **D**
119. **A**	120. **A**	121. **D**	122. **B**	123. **39**
124. **7**	125. **13**	126. **2929**	127. **2**	128. **3**

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129. **3**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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